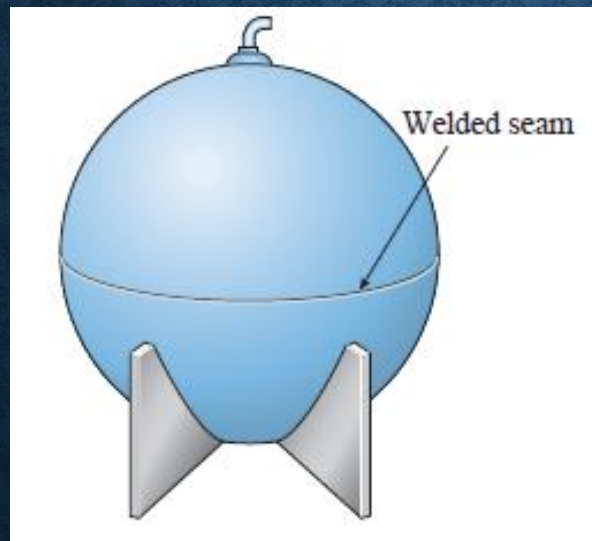


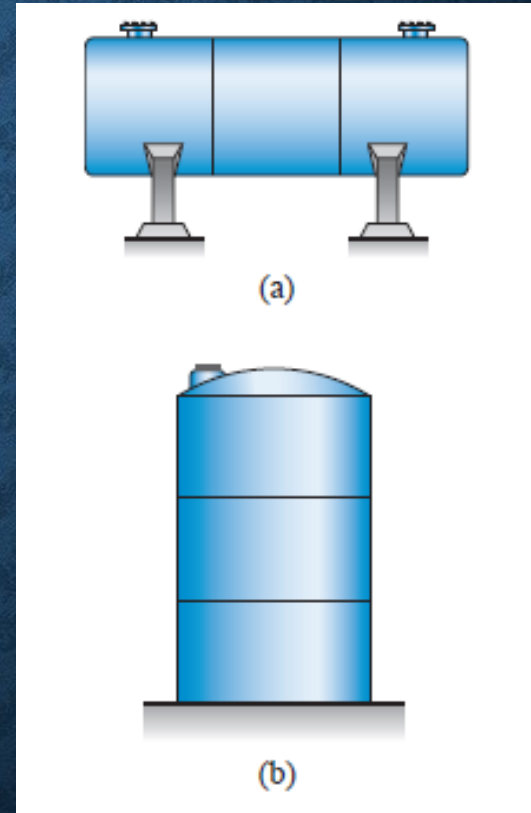
CHAPTER 6

THIN AND THICK WALL CYLINDERS

- A **Pressure vessel** is used to store fluid under a certain pressure. A pipe line through which pressurized fluid flows is treated as pressure vessel.
- Normally pressure vessels are of **Cylindrical** or **Spherical** shape.



(A)



(b)

• Figure 1: Pressure vessels (A) Spherical (B) Cylindrical

- Cylindrical pressure vessels with a circular cross section are found in many engineering application areas.
- These are some of the examples:
 - Compressed air tanks and rocket motors,
 - Fire extinguishers and spray cans, and
 - Propane tanks and grain silos.
 - Pressurized pipes, such as water-supply pipes and penstocks,
 - Boilers, and even
 - Soda bottles, are also classified as cylindrical pressure vessels.





- If the ratio of thickness to internal diameter of a cylindrical shell is less than about 1/20, the cylindrical shell is known as **Thin cylinder**.

$$\frac{d}{t} < 20 \text{ Thin Cylinder}$$

- If the ratio of thickness to internal diameter is more than 1/20, then cylindrical shell is known as **Thick cylinder**.

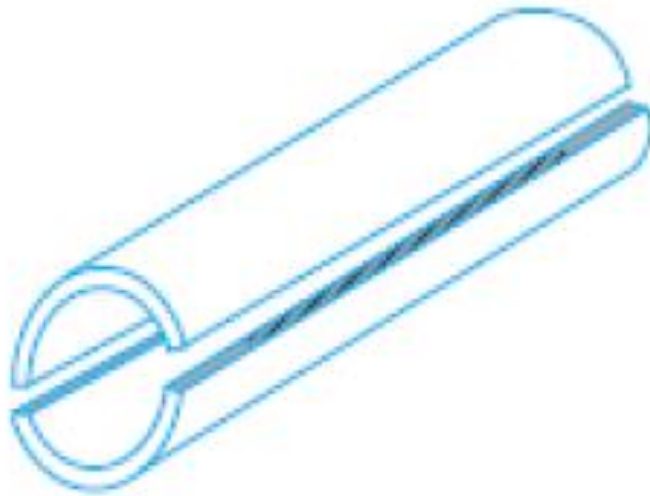
$$\frac{d}{t} > 20 \text{ Thick Cylinder}$$

The analysis of stresses induced in a thin cylindrical shell are made on the following assumptions:

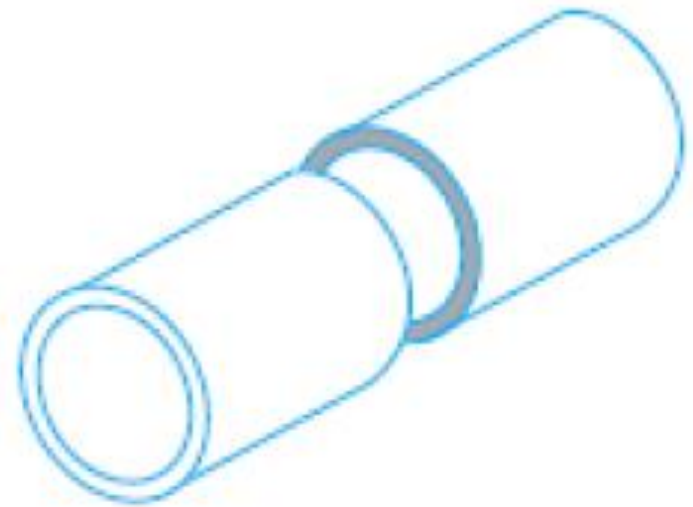
1. The effect of curvature of the cylinder wall is neglected.
2. The tensile stresses are uniformly distributed over the section of the walls.
3. The effect of the restraining action of the heads at the end of the pressure vessel is neglected.

When a thin cylindrical shell is subjected to an internal pressure, it is likely to fail in the following two ways:

1. It may fail along the longitudinal section (*i.e.* circumferentially) splitting the cylinder into two troughs.
2. It may fail across the transverse section (*i.e.* longitudinally) splitting the cylinder into two cylindrical shells.



(a) Failure of a cylindrical shell
along the longitudinal section.



(b) Failure of a cylindrical shell
along the transverse section.

7.1 THIN CYLINDERS UNDER INTERNAL PRESSURE

- When a thin-walled cylinder is subjected to internal pressure, three mutually perpendicular principal stresses will be set up within the wall of the cylinder,

Namely ;

1. Hoop or Circumferential Stress (σ_H)

This is directed along the tangent line to the circumference and tensile in nature. Thus, there will be increase in diameter.

2. Radial Stress (P_r)

It is compressive in nature. Its magnitude is equal to fluid pressure on the inside wall and zero on the outer wall if it is open to atmosphere.

3. Longitudinal Stress (σ_L)

This stress is directed along the length of the cylinder. This is also tensile in nature and tends to increase the length.

- Provided that the ratio of thickness to inside diameter of the cylinder is less than $1/20$, it is reasonable to assume that the hoop and longitudinal stresses are constant across the wall thickness
- And the magnitude of the radial stress set up is **so small** in comparison with the hoop and longitudinal stresses that it **can be neglected**.
- This is obviously an approximation since, in practice, it will vary from **zero at the outside surface** to a value equal to the **internal pressure at the inside surface**.
- For the purpose of derivation of stress formula it is also assumed that the ends of the cylinder and any riveted joints present have no effect on stresses produced.

7.1.1 Hoop or Circumferential Stress

- This is the stress which is set up in resisting the bursting effect of the applied pressure and can be most conveniently treated by considering the equilibrium of half of the cylinder as shown below in Figure 2.

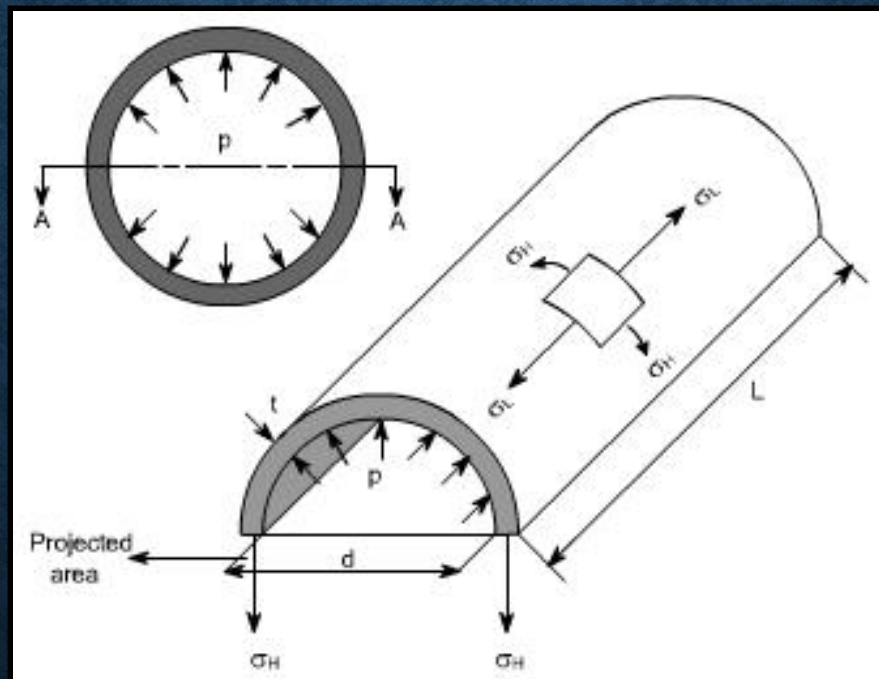


Figure 2: Half of a thin cylinder subjected to internal pressure

- Total force on half-cylinder due to internal pressure is :

$$P \times \text{Projected Area} = P \times dL$$

- Total resisting force due to hoop stress σ_H set up in the cylinder wall is :

$$2\sigma_H \times Lt$$

Therefore,

$$2\sigma_H \times Lt = P \times dL$$

Circumferential or Hoop Stress :

$$\sigma_H = \frac{Pd}{2t}$$

7.1.2 Longitudinal Stress

- Consider now the cylinder shown in Figure 3. Total force on the end of the cylinder owing to internal pressure

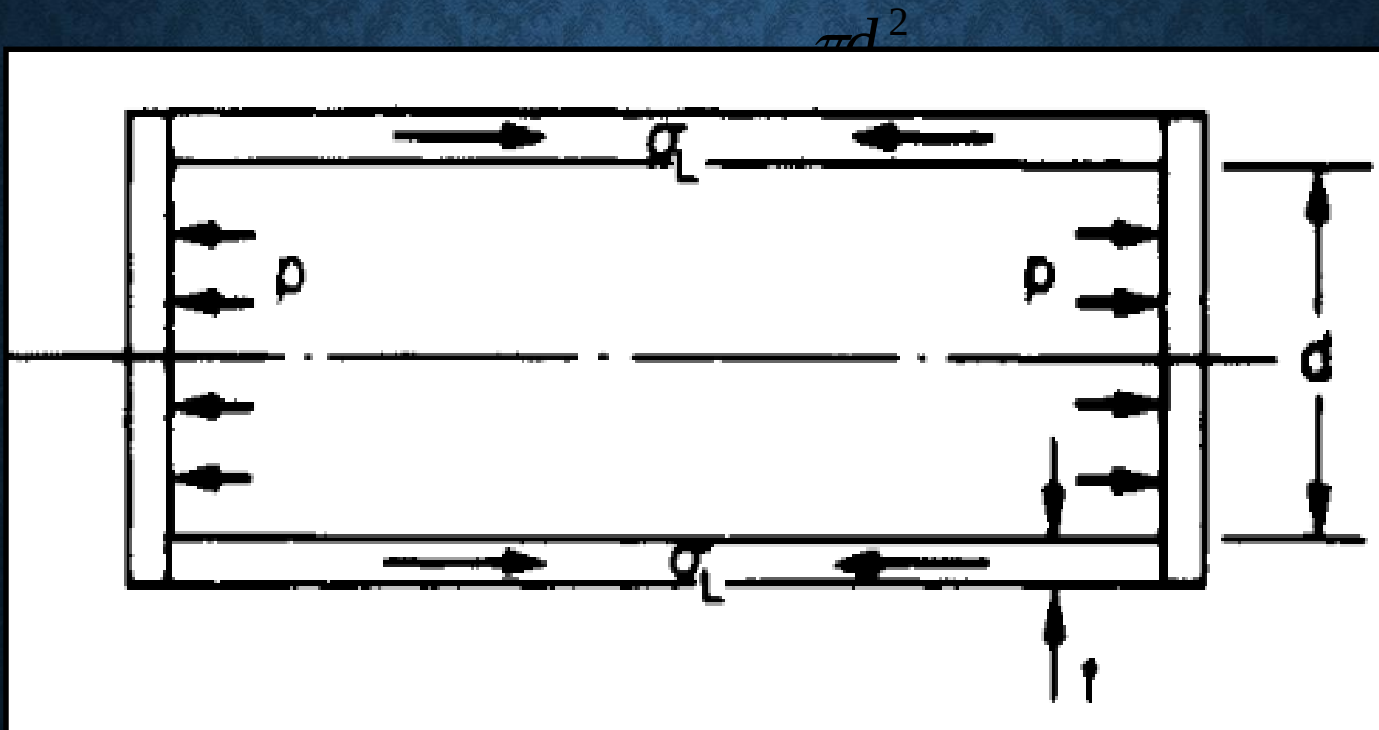


Figure 3: Cross-section of a thin cylinder

- Area of metal resisting this force = πdt (Approximately)

$$\text{Stress set-up} = \frac{\text{Force}}{\text{Area}} = P \times \frac{\pi d^2 / 4}{\pi dt} = \frac{Pd}{4t}$$

$$\text{Longitudinal Stress, } \sigma_L = \frac{Pd}{4t}$$

CHANGE IN DIMENSIONS

Change in Length

- The change in length of a thin cylinder can be determined from the longitudinal strain (we neglect the radial stress):

$$\text{Longitudinal strain} = \frac{1}{E} (\sigma_L - \nu \sigma_\theta)$$

Change in length, $\delta L = \text{longitudinal strain, } \epsilon_L \times \text{original length, } L$

$$\delta L = \frac{1}{E} (\sigma_L - \nu \sigma_\theta) L = \frac{1}{E} \left(\frac{p_i d}{4t} - \nu \frac{p_i d}{2t} \right) L$$

$$\delta L = \frac{p_i d}{4tE} (1 - 2\nu) L$$

Change in Diameter

- In the same way as above, we can determine the change in diameter from strain acting on the diameter, the *diametric* strain.

$$\text{diametral strain} = \frac{\text{change in diameter}}{\text{original diameter}}$$

We can find the change in diameter by considering the circumferential change. The stress acting around the circumference is the hoop or circumferential stress, σ_θ giving rise to the circumferential strain ε_θ .

$$\text{change in circumference} = \text{strain} \times \text{original circumference}$$

$$\delta C = \varepsilon_\theta \times \pi d$$

The new circumference is given by:

$$\text{new circumference} = \pi d + \pi d \varepsilon_{\theta} = \pi d(1 + \varepsilon_{\theta})$$

And it can be seen that this describes the circumference of a circle of diameter $d(1 + \varepsilon_{\theta})$.

$$\text{new diameter} = d(1 + \varepsilon_{\theta})$$

$$\text{change in diameter} = d \varepsilon_{\theta}$$

Therefore:

$$\text{diametral strain} = \frac{d \varepsilon_{\theta}}{d} = \varepsilon_{\theta}$$

Thus demonstrating that the diametric strain is equal to the hoop strain.

Therefore:

$$\text{change in diameter, } \delta d = d \varepsilon_{\theta} = \frac{d}{E} (\sigma_{\theta} - \nu \sigma_L) = \frac{d}{E} \left(\frac{p_i d}{2t} - \nu \frac{p_i d}{4t} \right)$$

$$\delta d = \frac{p_i d^2}{4tE} (2 - \nu)$$

CHANGE IN VOLUME

- If a fluid is used as the pressurization medium the fluid itself will change in volume as pressure is increased and this must be taken into account when calculating the amount of fluid which must be pumped into the cylinder in order to raise the pressure by a specified amount, the cylinder being initially full of fluid at atmospheric pressure.

$$\text{Bulk modulus } K = \frac{\text{Volumetric Stress}}{\text{Volumetric Strain}}$$

- Volumetric Stress = Pressure(P)

$$\text{Volumetric Strain} = \frac{\text{Change in Volume}}{\text{Original Volume}} = \frac{\delta V}{V}$$

$$K = \frac{P}{\delta V / V} = \frac{PV}{\delta V}$$

- Change in volume of fluid under pressure= PV/K
- The extra fluid required to raise the pressure must, therefore, take up this volume together with the increase in internal volume of the cylinder itself.
- Extra Fluid required to raise cylinder pressure by P

$$= \frac{Pd}{4tE} [5 - 4\nu] V + \frac{PV}{K}$$

- OR, $\text{change in volume} = \text{volumetric strain} \times \text{original volume}$

Given that:

volumetric strain = sum of three mutually perpendicular direct strains

$$\text{volumetric strain} = \varepsilon_L + 2\varepsilon_\theta = \frac{1}{E}(\sigma_L - \nu\sigma_\theta) + \frac{2}{E}(\sigma_\theta - \nu\sigma_L)$$

$$\text{volumetric strain} = \frac{1}{E}(\sigma_L - \nu\sigma_\theta + 2\sigma_\theta - 2\nu\sigma_L) = \frac{1}{E}(\sigma_L + 2\sigma_\theta - \nu(\sigma_\theta + 2\sigma_L))$$

$$\text{volumetric strain} = \frac{1}{E} \left(\frac{p_i d}{4t} + 2 \frac{p_i d}{2t} - \nu \left(\frac{p_i d}{2t} + 2 \frac{p_i d}{4t} \right) \right) = \frac{p_i d}{4tE} (5 - 4\nu)$$

Therefore:

$$\delta V = \frac{p_i d}{4tE} (5 - 4\nu) V$$

CYLINDRICAL VESSEL WITH HEMISPHERICAL ENDS

- Consider now the vessel shown below in which the wall thickness of the cylindrical and hemispherical portions may be different (this is sometimes necessary since the hoop stress in the cylinder is twice that in a sphere of the same radius and wall thickness). For the purpose of the calculation the internal diameter of both portions is assumed equal.

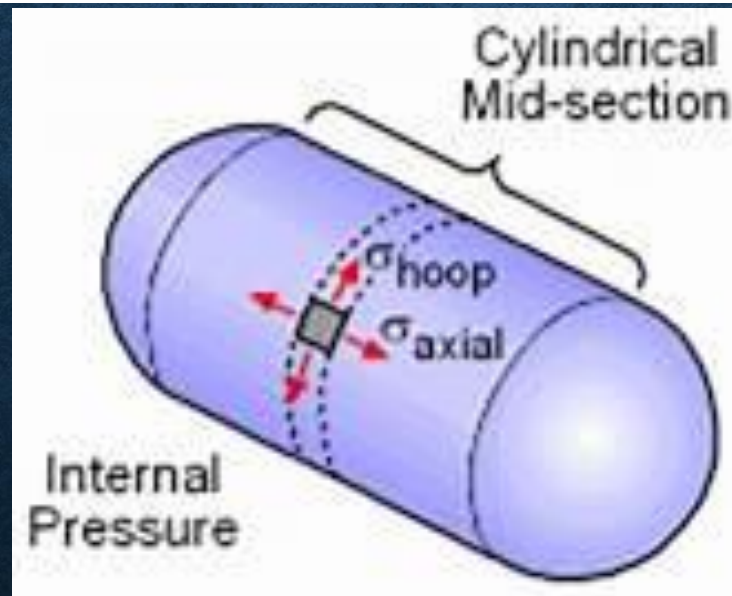
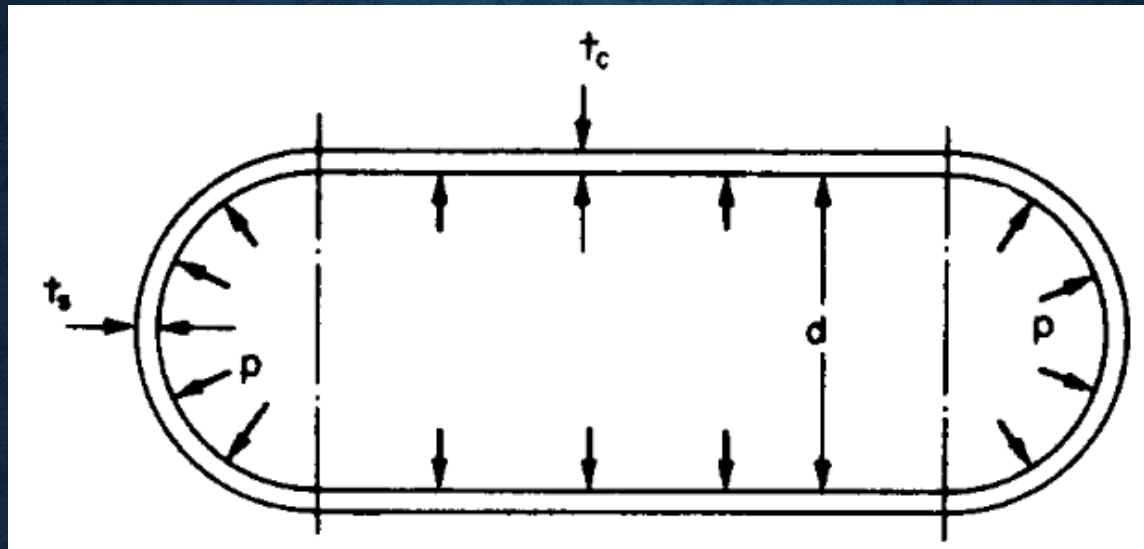


Figure 4: Cross-Section of a thin cylinder with hemispherical ends

a) For the cylindrical Portion

$$\sigma_{H_C} = \frac{Pd}{2t_C} \quad \text{Hoop or Circumferential Stress}$$

$$\sigma_{L_C} = \frac{Pd}{4t_C} \quad \text{Longitudinal Stress}$$

$$\begin{aligned} \text{Hoop or circumferential Strain} &= \frac{1}{E} [\sigma_{H_C} - \nu \sigma_{L_C}] \\ &= \frac{Pd}{4t_C E} [2 - \nu] \end{aligned}$$

b) For the Hemispherical Ends

$$\text{Hoop Stress} = \sigma_{H_s} = \frac{Pd}{4t_s}$$

$$\text{Hoop Strain} = \frac{1}{E} [\sigma_{H_s} - \nu \sigma_{H_s}] = \frac{Pd}{4t_s E} [1 - \nu]$$

Equating the two strains in order that there shall be no distortion of the Junction,

$$\frac{Pd}{4t_c E} [2 - \nu] = \frac{Pd}{4t_s E} [1 - \nu]$$

$$\frac{t_s}{t_c} = \frac{(1 - \nu)}{(2 - \nu)} \cong \frac{0.7}{1.7} \text{ (Poisson's Ratio for general steel work of 0.3)}$$

Conclusion: The thickness of the cylinder walls must be approximately **2.4 times** that of the hemispherical ends for no distortion of the junction to occur.

Example 2:

A thin cylinder 60mm internal diameter, 225mm long with walls 2.7mm thick is subjected to an internal pressure of 6MN/m². You may assume that $G = 200$ MPa and $\nu = 0.3$. Calculate:

- A. The hoop stress
- B. The longitudinal stress
- C. The change in length
- D. The change in diameter

THICK CYLINDERS

DIFFERENCE IN TREATMENT BETWEEN THIN AND THICK CYLINDERS-BASIC ASSUMPTIONS

- The theoretical treatment of thin cylinders assumes that **the hoop stress is constant across the thickness of the cylinder wall**, and also that **there is no pressure gradient across the wall**.
- However, Neither of these assumptions can be used for thick cylinders for which the variation of hoop and radial stresses is shown in figure below, their values being given by the **LAME EQUATIONS**:

$$\sigma_H = A + \frac{B}{r^2} \quad \text{and} \quad \sigma_r = A - \frac{B}{r^2}$$

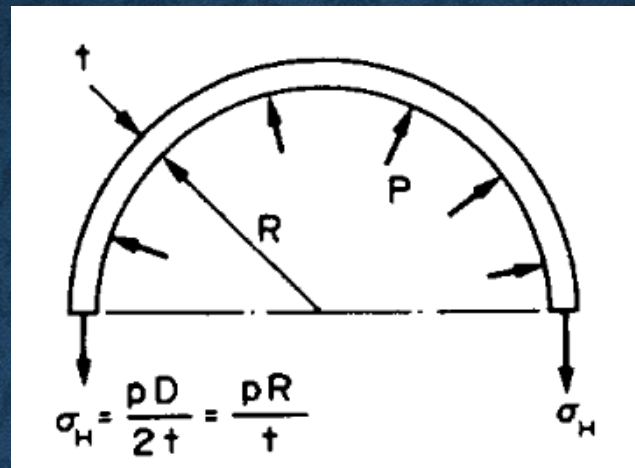


Figure 5: Thin cylinder subjected to internal pressure

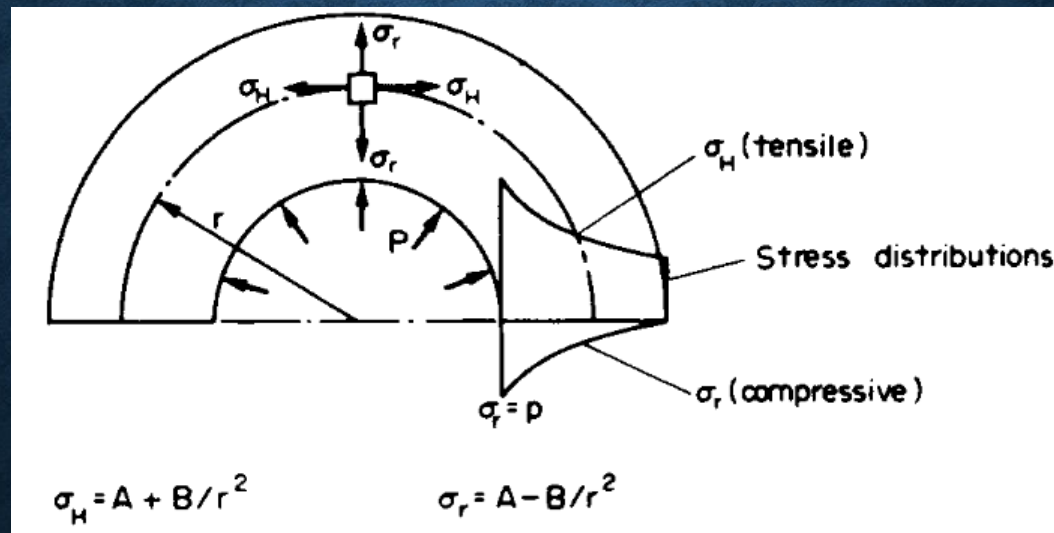
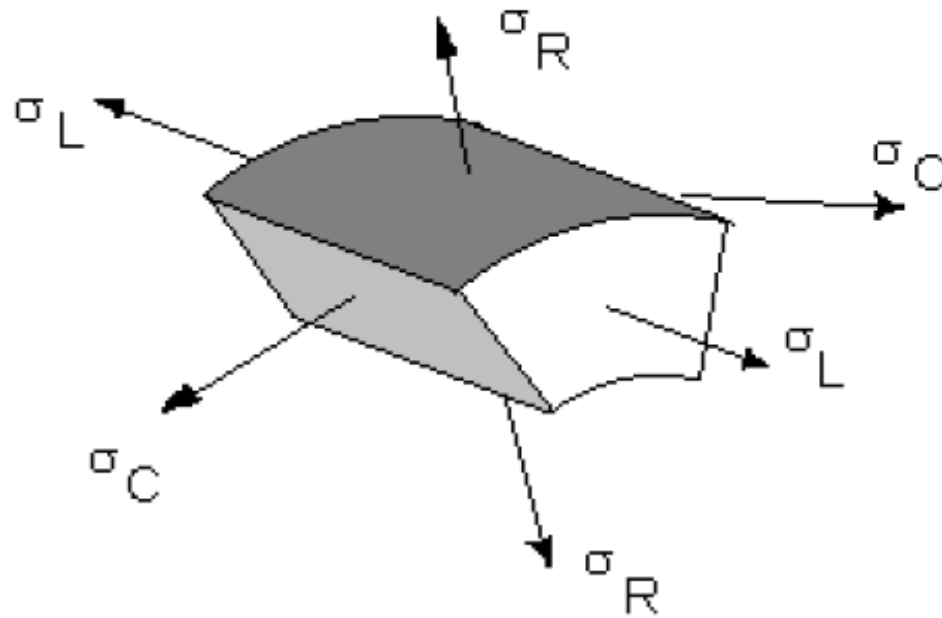


Figure 6: Thick cylinder subjected to internal pressure

- Development of the theory for thick cylinders is concerned with sections remote from the ends since distribution of the stresses around the joints makes the analysis at the ends particularly complex.
- For central sections the applied pressure system which is normally applied to thick cylinders is symmetrical, and all points on an element of the cylinder wall will be displaced by the same amount, this amount depends on the radius of the element.

Consider a small section of the wall.



σ_L = Longitudinal stress

σ_R = Radial stress

σ_C = Circumferential stress

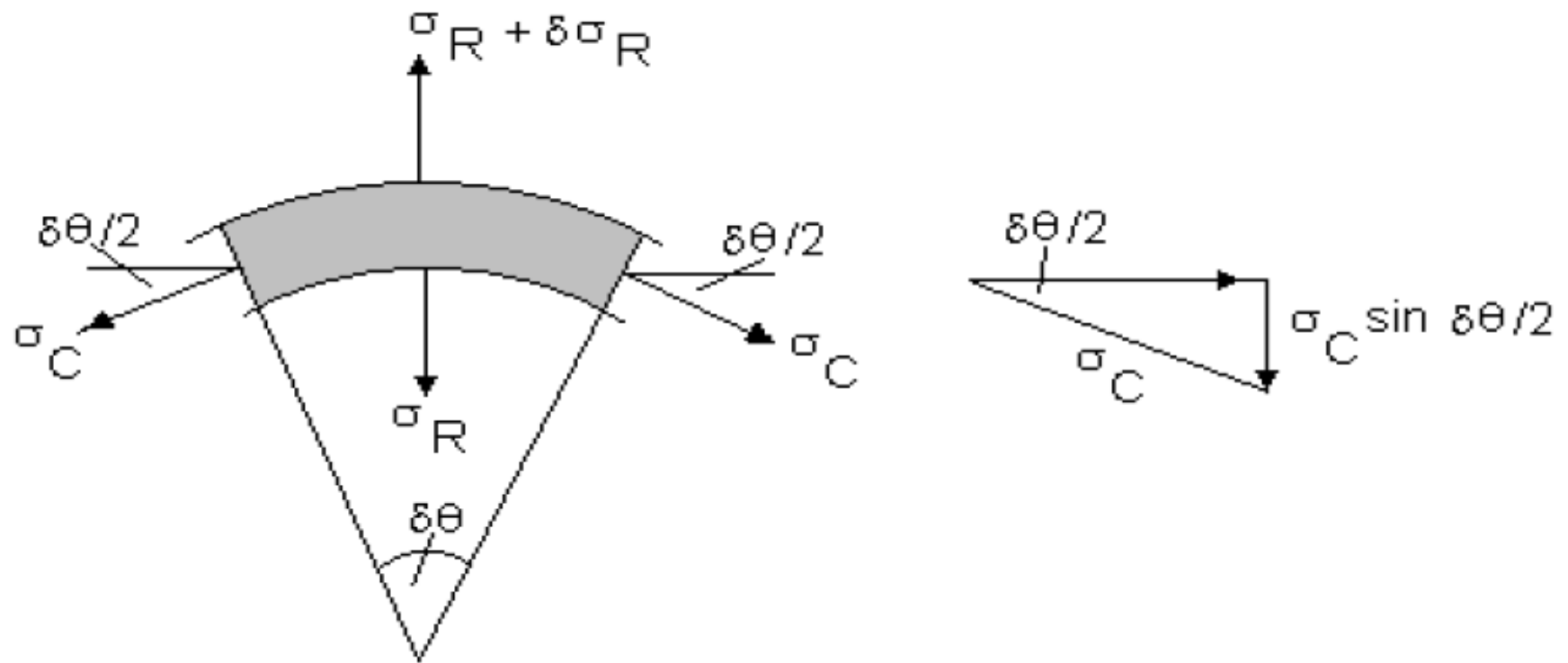
We have 3 stresses in mutually perpendicular directions, the corresponding strains are

$$\epsilon_L = \frac{1}{E} \{ \sigma_L - \nu(\sigma_R + \sigma_C) \}$$

$$\epsilon_C = \frac{1}{E} \{ \sigma_C - \nu(\sigma_L + \sigma_R) \}$$

$$\epsilon_R = \frac{1}{E} \{ \sigma_R - \nu(\sigma_C + \sigma_L) \}$$

Next consider the forces acting on a section of the wall.



Balance the forces vertically (assuming 1 m of length).

Remember the length of an arc is radius $\times$ angle

The area of the top curved surface is $(r + \delta r)\delta\theta \times 1$

The area of the bottom curved surface is $r\delta\theta \times 1$

Remember Force is stress $\times$ area.

The vertical force up is $(\sigma_R + \delta\sigma_R)(r + \delta r)\delta\theta$

The vertical force down is $\sigma_R r \delta\theta + 2 \sigma_C \delta r \sin\delta\theta/2$

Remember for small angles the sin is the same as the angle in radians.

$$\sin \delta\theta/2 = \delta\theta/2$$

Balancing the forces we have

$$(\sigma_R + \delta\sigma_R)(r + \delta r)\delta\theta = \sigma_R r \delta\theta + 2\sigma_C \delta r \frac{\delta\theta}{2}$$

This resolves down to

$$\frac{\delta\sigma_R}{\delta r} = \sigma_C - \sigma_R$$

In the limit this becomes

$$\frac{d\sigma_R}{dr} = \sigma_C - \sigma_R \dots\dots\dots(4.1)$$

The strain is given by

$$\epsilon_L = \frac{1}{E} \{ \sigma_L - \nu(\sigma_C + \sigma_C) \}$$

Since ϵ_L and σ_L are constant then it follows that $(\sigma_R + \sigma_C) = \text{constant}$.

The solution is simplified by making the constant $2a$

$$\begin{aligned} (\sigma_R + \sigma_C) &= 2a \\ \sigma_C &= 2a - \sigma_R \end{aligned} \dots\dots\dots(4.2)$$

Substitute (4.2) into (4.1) and

$$\frac{r d\sigma_R}{dr} = 2a - \sigma_R - \sigma_R = 2a - 2\sigma_R$$

multiply all by r and rearrange

$$\frac{r^2 d\sigma_R}{dr} - 2ar = 2r\sigma_R$$

It can be shown that

$$\frac{d(r^2 \sigma_R)}{dr} = \frac{r^2 d\sigma_R}{dr} + 2r\sigma_R$$

$$\frac{d(r^2 \sigma_R)}{dr} = 2ar$$

$$(r^2 \sigma_R) = \int 2ar dr = ar^2 - b$$

where b is a constant of integration.

$$\sigma_R = a - \frac{b}{r^2}$$

$$\sigma_C = a + \frac{b}{r^2}$$

In order to solve problems, the constants a and b must be found from boundary conditions.

Remember: a boundary condition is a known answer such as knowing what the pressure or stress is at a given radius.

When atmospheric pressure acts on one side of the wall, it is best to use gauge pressure in the calculations. This makes atmospheric pressure zero and all other pressures are relative to it.

Remember: absolute pressure = gauge pressure + atmospheric pressure.

THICK CYLINDER-INTERNAL PRESSURE ONLY

- Consider now the thick cylinder shown in Figure 9 subjected to an internal pressure P , the external pressure being zero.

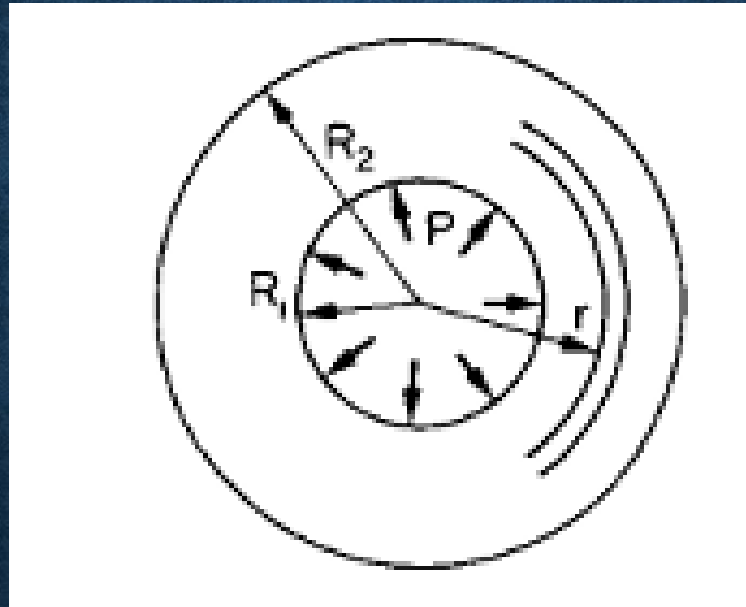


Figure 9: Cylinder cross section

- The two known conditions of stress which enable the Lamé constants A and B to be determined are:

At $r = R_1$ $\sigma_r = -P$ and at $r = R_2$ $\sigma_r = 0$

N.B: The internal pressure is considered as a radial stress since it will produce a radial compression of the cylinder walls and the normal stress convention takes compression as **negative**.

Taking the above conditions in Lamé's equation gives

$$-P = A - \frac{B}{R_1^2}$$

$$0 = A - \frac{B}{R_2^2}$$

$$\text{i.e. } A = \frac{PR_1^2}{(R_2^2 - R_1^2)} \text{ and } B = \frac{PR_1^2 R_2^2}{(R_2^2 - R_1^2)}$$

- Radial stress

$$= -P \left[\frac{(R_2 / r)^2 - 1}{k^2 - 1} \right]$$

Where k is the diameter ratio = R_2/R_1

- Hoop stress

$$= P \left[\frac{(R_2 / r)^2 + 1}{k^2 - 1} \right]$$

- Longitudinal stress with external pressure P_2

$$= \frac{P_1 R_1^2 - P_2 R_2^2}{R_2^2 - R_1^2}$$

- Where P_1 is internal pressure and P_2 to be external pressure

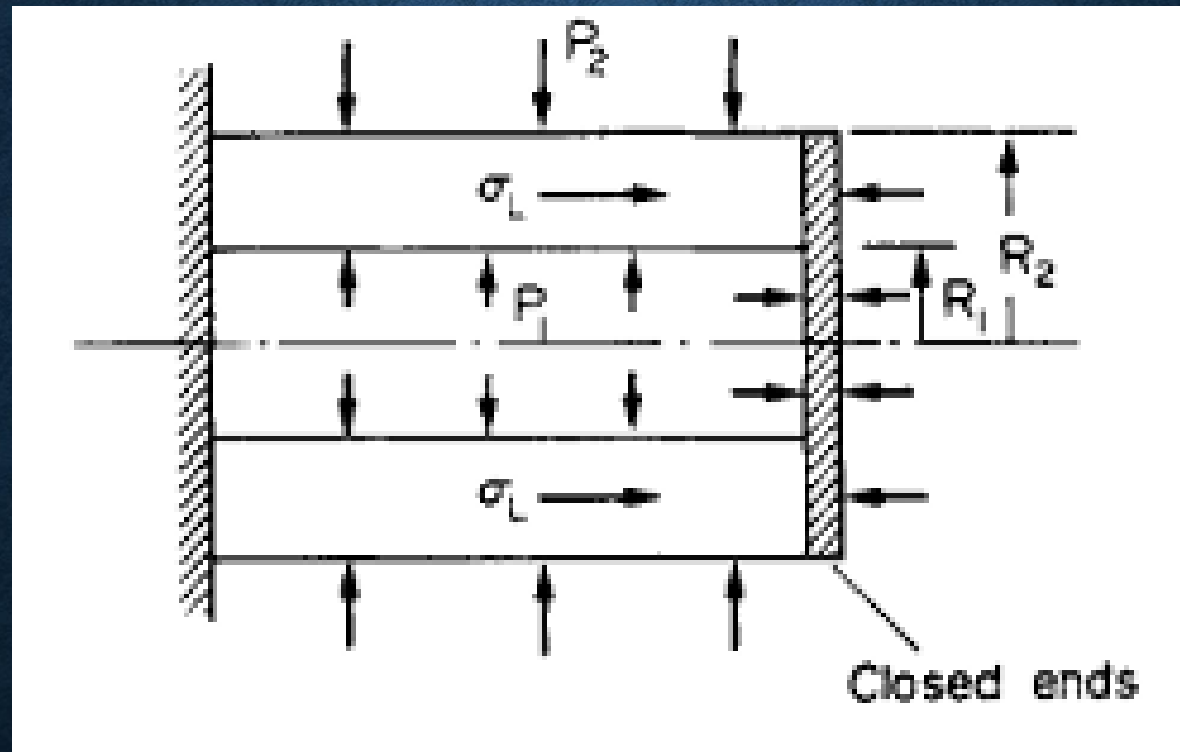


Figure 10: Cylinder longitudinal section

END